

**A Project Report
On
BidSell (Online Auction Platform)**

By

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**A Project Submitted in partial fulfillment of the requirements for the degree of
Bachelor of Science in Computer Science and Engineering (CSE)**

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Declaration

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This is to certify that the project work submitted by **Al-Rakib, Md. Ruman, Amir Kashru and Shah Md. Golam Morshed** entitled “**BidSell**” has been approved by the project Review Committee partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Science and Engineering (CSE) in the Department of Computer Science and Engineering (CSE), University of Information Technology and Sciences (UITs), Dhaka, Bangladesh in **December, 2024**.

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Abstract

BidSell is an online auction platform designed to address the gap in secure, user-friendly, and dynamic bidding systems in the Bangladeshi market. The platform provides buyers and sellers with a seamless way to exchange high-value goods in a secure and transparent environment. Featuring three distinct panels—Buyer, Seller, and Admin—BidSell allows sellers to list products for auction, buyers to place competitive bids, and administrators to oversee and regulate the entire ecosystem.

The motivation behind this project stems from the lack of accessible and efficient auction systems in Bangladesh, coupled with the need for platforms that cater to local requirements such as mobile compatibility, secure transactions, and intuitive user interfaces. Existing systems like AdminDeal and Slibuy fail to offer real-time bidding, adequate user verification, and mobile-optimized experiences. Similarly, platforms like Bikroy.com focus solely on direct transactions, leaving the auction segment largely unaddressed. This inspired the development of BidSell, aiming to transform the auction process and bridge these gaps.

The problem identified includes trust issues, static bidding processes, and non-intuitive designs in existing platforms, which deter participation and limit user engagement. BidSell addresses these limitations by implementing a robust verification process, real-time bidding updates, and a modern user interface built with technologies like Java, Firebase Firestore, and Firebase Authentication.

The project follows an Agile development methodology to ensure flexibility, user-centric design, and incremental improvements. Functional requirements include user authentication, category management, auction creation, real-time notifications, and secure payment processing. Non-functional requirements such as scalability, reliability, and usability are emphasized to provide a seamless and efficient experience. The platform integrates cutting-edge features like AI-based fraud detection (proposed for future work), interactive dashboards, and automated bidding systems.

The outcome of this project is a fully functional auction platform that enables transparent and efficient transactions. Sellers can easily list their products, buyers can participate in real-time bidding, and administrators can manage the system effectively. The use of scalable technologies and secure payment systems ensures reliability, while real-time updates and dynamic features enhance user engagement.

In conclusion, BidSell not only fills a critical void in the Bangladeshi e-commerce landscape but also contributes to the digital transformation of the region by promoting trust, accessibility, and economic growth. This platform has the potential to empower individuals and businesses while fostering a collaborative and competitive online marketplace. Future enhancements, such as AI-driven fraud detection, personalized recommendations, and web application development, will further expand its capabilities and reach.

Keywords: Online Auction Platform, Buyer Panel, Seller Panel, Admin Panel, User Registration, User Verification, Product Listing, Real-Time Bidding, Secure Transactions.

Preface

This B.Sc. capstone project is outlined based on the results obtained from the laboratory experiment. This is carried out in the Department of Computer Science and Engineering (CSE), Faculty of Science and Engineering, at University of Information Technology and Sciences (UITS), Dhaka, Bangladesh.

This capstone project includes 5 chapters which are briefed as follows:

Chapter-1

Chapter 1 provides a basic introduction.

Chapter-2

Chapter 2 provides a background study.

Chapter-3

Chapter 3 provides the methodology provides the methodology you applied. This chapter contain flowcharts, algorithms etc.

Chapter-4

Chapter 4 provides implementation of our work

Chapter-5

Chapter 5 provides conclusion and future work.

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Abbreviations and Symbols

CSE	Computer Science and Engineering
OS	Operating System
RAD	Rapid Application Development
DFD	Data Flow Diagram
ER	Entity-Relationship Diagram
UML	Unified Modeling Language
UI	User Interface
OTP	One-Time Password
API	Application Programming Interface
XML	Extensible Markup Language
NoSQL	Not Only SQL (database type)
AI	Artificial Intelligence
PSP	Payment Service Provider

Chapter 1

Introduction

1.1 Introduction

BidSell is an online auction platform that provides a secure and open environment for buying and selling high-value goods. Catering to both buyers and sellers through dedicated panels, it streamlines the auction process by allowing sellers to easily list their expensive and valuable products and buyers to bid on items of interest within specified time frames. The platform also features an Admin panel, enabling administrators to oversee the entire bidding system, manage registered users, and approve products by category to ensure a smooth and regulated marketplace. By leveraging the dynamic nature of online auction systems, BidSell connects individuals looking to exchange high-value items in a trustworthy and user-friendly setting.

1.2 Motivation

Online auctions are not widely used in Bangladesh, and there is a lack of user-friendly apps that facilitate participation for everyone. We observed this and desired to make a modification. This is the reason we are developing a new online bidding platform that is easy to use and available to all individuals in Bangladesh. Our application enables users to easily post items for sale and make offers on different products. By making the auction process simpler, we are providing people with a secure and easy method to buy and sell goods. We are excited to transform the auction sector in Bangladesh to make it more convenient and available to all.

1.3 Goals of the Project

Our goal is to transform the buying and selling of goods and services in Bangladesh by developing an auction app that is easy to use for the people in the market. Acknowledging the absence of user-friendly auction platforms, our goal is to offer a secure online space where users can engage in auctions, bid confidently, and transact smoothly. Our app will meet the specific needs of Bangladeshi users by including unique features and using a culturally relevant business model, which will help create a lively online marketplace. Our dedication is to improve user experience, support transparency, and empower individuals and businesses to participate in successful and reliable auction activities. This will contribute to the economic development and digital modernization of our country.

1.4 Why we select this project

We have decided to create a customized online auction system for the Bangladeshi market for several compelling reasons as our Capstone Project. Firstly, the e-commerce industry in Bangladesh is growing rapidly, yet there is a noticeable lack of auction platforms that are both user-friendly and tailored to the specific needs and preferences of the local population. While platforms like AdminDeal[1] and Slibuy[2] offer auction services, they suffer from critical limitations such as the absence of real-time bidding and inadequate user verification systems, leading to trust issues and suboptimal user experiences.

Our project seeks to address these gaps by developing an auction platform that integrates robust real-time bidding features, mandatory user verification processes, and a smooth, intuitive interface optimized for mobile users. This approach aims to establish a trustworthy and dynamic online marketplace that encourages participation and engagement.

Additionally, building an internet-based auction system is technically challenging, requiring expertise in secure payment processing, scalable infrastructure, and live bidding features. Tackling these challenges will allow us to enhance our programming and software development skills significantly.

Furthermore, the opportunity to innovate in this sector is immense. Incorporating features such as localized payment options, multilingual support, and mobile accessibility will ensure that our platform caters specifically to the cultural and economic context of Bangladesh. By addressing these needs, our project not only fills a critical gap in the market but also contributes to the digital transformation of the country's e-commerce landscape.

1.5 Conclusion

BidSell aims to fill the gap of user-unfriendly online auction platforms in Bangladesh by offering a secure, transparent, and easily accessible marketplace for valuable goods. The platform allows users to purchase and sell with confidence by streamlining the auction process and including features designed for local requirements. This project aims to boost economic growth and improve our technical abilities, helping to modernize Bangladesh.

Chapter 2

Background Studies

2.1 Introduction

The rapid growth of e-commerce platforms has revolutionized the way people buy and sell products. Auction apps, which allow users to bid in real-time on items of interest, are playing a key role in this digital transformation. Platforms such as AdminDeal [1] and Slibuy [2] play an important role in the auction space but face limitations that leave room for innovation. Similarly, e-commerce platforms such as Bikroy.com address a different segment of the online market by focusing on direct transactions rather than bidding. These existing systems will serve as a benchmark to understand the current situation, identify gaps, and suggest improvements through our proposed auction application.

2.2 Related Work

2.2.1 AdminDeal and Slibuy:

For this project we study about AdminDeal [1] and Slibuy [2]. Both are online auction platforms that facilitate the buying and selling of products through bidding mechanisms. We also study about Bikroy.com [3] this is a most popular online platform where buyer can buy product and seller can sell their product without bidding mechanisms.

2.3 Challenges

2.3.1 Main Challenge of AdminDeal:

- **No Verification Option:** Lack of user verification. It doesn't have proper verification system that can identify a person. So it leads to trust issues, such as fake bidding and unreliable seller trustworthiness.
- **No Real-time Bidding:** It has absence of a dynamic bidding process limits the excitement and competitive experience of live auctions.

2.3.2 Main Challenge of Slibuy:

- **Lower User Responsiveness:** It has a big issue that is poor user interface (UI) and inadequate real-time notifications result in a suboptimal user experience.

2.4 Bikroy.com

Although Bikroy.com[3] is not an auction platform, it provides insights into the e-commerce landscape where direct transactions occur between buyers and sellers.

2.4.1 Main Challenge of Bikroy.com:

- **Price Bargaining:** The lack of structured pricing negotiation processes leads to inefficient transaction methods among users.

2.5 Limitations of Existing Systems

- **Trust Issues:** Most of auction platform has lack of user verification system like AdminDeal and Slibuy. As a result, it is hard to identify a person.
- **Static Bidding:** Most of the auction platforms has operate without real-time bidding.
- **Transparency:** Lack of transparency in bidding processes or hidden fees can lead to mistrust among users and deter participation
- **Mobile Compatibility:** Some existing systems might not have optimized interfaces for mobile devices, limiting accessibility and convenience for users
- **Non-Intuitive User Interfaces:** User-centric designs are frequently neglected by the majority of execution platforms, which results in poor responsiveness and decreased satisfaction levels.

2.6 Feasibility Study

The feasibility study aims to evaluate the practicality of implementing the proposed auction app, focusing on technical, economic, and operational aspects.

2.6.1 Technical Feasibility

The technical feasibility of the proposed auction app evaluates its implementation using a robust set of tools, technologies, and dependencies. The selected tools ensure scalability, efficiency, and a seamless user experience.

2.6.2 Core Technologies

1. **Programming Language:**
 - **Java:** Will be used to control the business logic of the application, ensuring a robust and maintainable backend system.
2. **Design Language:**
 - **XML:** For creating the front-end layouts and interfaces, ensuring a clean and responsive user experience.

3. **Database:**
 - **Firebase FireStore:** A NoSQL cloud database that supports real-time synchronization, which is essential for dynamic updates during auctions.
4. **User Verification:**
 - **Firebase Authentication:** Provides a secure and reliable user authentication system, supporting methods like email, phone, and social media logins.
5. **Design Framework:**
 - **Material Design:** Ensures a modern, intuitive, and consistent user interface for both buyers and sellers.

2.6.3 Dependencies

To streamline development and enhance functionality, the following dependencies will be used:

1. **Circle Image View:**
 - To create rounded profile pictures for a visually appealing user experience.
2. **Image Slider Show:**
 - To implement smooth image sliders for showcasing auction items.
3. **Glide:**
 - To efficiently load and display images from the internet, ensuring optimal performance and user experience.
4. **Payment Gateway Integration:**
 - To handle secure online transactions, ensuring trust and seamless payments between buyers and sellers.
5. **OkHTTP3:**
 - To handle network requests and ensure reliable communication with external APIs.
6. **Lottie Animation:**
 - To add engaging animations to improve the app's user interface and interactivity.

2.6.4 Integration Strategy

1. **Real-Time Bidding:**
 - Utilizing Firebase Firestore's real-time synchronization capabilities to update bid statuses dynamically across all devices.
2. **User Verification:**
 - Firebase Authentication will securely verify user identities and mitigate fraudulent activities.
3. **Interactive Design:**
 - Material Design combined with Lottie animations will ensure an appealing and responsive interface.
4. **Secure Transactions:**

- Uddoktapay will provide a robust and trustworthy payment solution, ensuring safe transactions.
5. **Media Management:**
- Glide and ImageSliderShow will handle images effectively, enhancing item visualization.

2.7 Proposed Features of the New Auction App

1. User Verification:

- Mandatory identity verification for both buyers and sellers.
- Integration of email and phone number verification.

2. Real-Time Bidding:

- Real-Time bidding updates for dynamic user interaction.

3. Enhanced User Interface:

- A modern and responsive design for better accessibility.
- Easy-to-navigate dashboards for buyers and sellers.

4. Secure Transactions:

- Integration with secure payment gateways to handle financial transactions.

2.8 Economic Feasibility

The proposed auction app is economically feasible, offering significant benefits to individuals, small businesses, sellers, and the broader community. Below is an analysis based on the identified opportunities:

2.8.1 Opportunities for Individuals or Small Businesses:

- Provides a platform for small businesses and individuals to reach a wider audience.
- Enables cost-effective selling without the need for physical stores

2.8.2 Empower Sellers:

- Sellers can list products quickly with minimal overhead.
- Real-time bidding encourages competitive pricing and maximizes revenue potential.

2.8.3 Economic Growth:

- Facilitates seamless transactions, boosting e-commerce activities.
- Encourages local and global trade, contributing to overall economic development.

2.8.4 Community among Buyers and Sellers:

- Builds trust with verified profiles and transparent transactions.
- Fosters a collaborative ecosystem for sustainable commerce.
- Encourages repeat usage, driving long-term profitability.

2.9 Revenue Model

The proposed auction app generates profit through multiple monetization strategies:

1. Advertising

- **Banner Ads:** Display targeted ads within the app for products or services relevant to users.
- **Promoted Listings:** Allow sellers to pay for higher visibility of their auction items.
- **Sponsored Categories:** Feature brands or sellers in specific categories to enhance their reach.

1. Commission from Seller

- **For Seller:** Specific commission will be deducted from the seller payment.

3. Additional Features for Buyers

- **Bidding Boost:** Offer buyers the option to pay for automated bidding or notifications on desired items.

2.10 Conclusion

The proposed auction app addresses the limitations of existing platforms by focusing on trust, real-time interaction, and user-centric design. By utilizing advanced technologies and implementing robust features, the app aims to set a new standard for online auction systems, ensuring a seamless experience for both buyers and sellers. This innovative approach will position the app as a leader in the competitive market of online auctions.

Chapter 3

Methodology

3.1 Introduction

In software development, methodologies provide guidelines, rules, and principles that developers follow to ensure systematic progress. Each methodology follows a unique approach, and the selection depends on the project requirements and constraints. Below, the methodologies are described and tailored for the auction app.

3.2 Agile Model

The **Agile Model** prioritizes collaboration, customer feedback, and adaptability. It divides the project into short development cycles called sprints, promoting continuous improvement. Agile software development is an iterative and flexible approach, emphasizing collaboration, customer feedback, and adaptability. It is particularly effective for complex projects where requirements evolve dynamically.

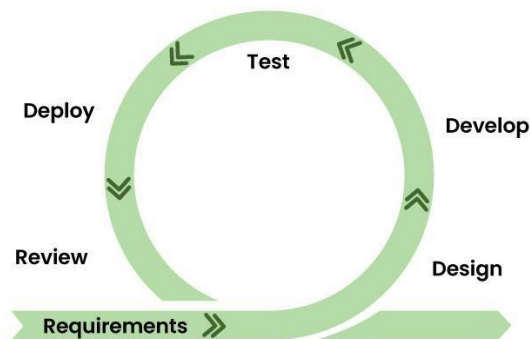


Figure 3.1: Agile Model [4]

3.2.1 Advantages of Agile Model

- In this, two programmers work together due to which the code is error free and there are very few mistakes in it.
- In this the software project is completed in a very short time.
- In this the customer representative has an idea of each iteration so that he can easily change the requirement.
- This is a very realistic approach to software development.
- In this, focus is given on teamwork.
- There are very few rules in this and documentation is also negligible.
- There is no need for planning in this.
- It can be managed easily.
- It provides flexibility to developers.

3.2.2 Key Principles of Agile:

- Adaptive planning and incremental delivery.
- Continuous improvement through frequent user feedback.
- Collaboration between cross-functional teams.
- Rapid and flexible responses to changes in requirements.

3.2.3 Agile for Auction App:

- Divides the project into manageable sprints, focusing on high-priority features like registration, bidding.
- Regular reviews after each sprint ensure alignment with user expectations.
- Accommodates changes, such as integrating advanced payment options or enhancing real-time bidding features.

3.3 Why Have We Chosen Agile?

The Agile methodology offers several advantages for the auction app:

1. **Flexibility:**
 - Agile accommodates changes in requirements, such as adding new features or modifying existing ones based on user feedback.
2. **User-Centric Development:**
 - Frequent feedback loops ensure that the app aligns with the needs of buyers and sellers.
3. **Cost and Time Efficiency:**
 - Testing and approval occur at each sprint, reducing the risk of expensive fixes later in development.
4. **Risk Mitigation:**
 - Allows for early detection and resolution of issues, ensuring smooth progress.
5. **Continuous Improvement:**
 - Regular updates and refinements enhance the app's functionality and usability.
6. **Collaborative Approach:**
 - Encourages collaboration between developers, designers, and stakeholders for better results.

3.4 Functional Requirements

3.4.1. Login Module

- **Features:**
 - User Registration: Allows new users to create an account (Admin, Seller, Bidder).
 - Authentication: Secure login for users using Firebase Authentication.
 - Change Password: Users can update their password after login.
 - Forgot Password: Enables users to reset their password via email or OTP.

3.4.2. Category Management

- **Features:**
 - Admin can create, update, and delete categories.
 - Categories help organize products for easy browsing and searching.

3.4.3. Search Module

- **Features:**
 - Category-wise search to filter items.
 - Keyword-based search for quick access to auction items.

3.4.4. Auction Module

- **Features:**
 - Sellers can upload products for auction with images, descriptions, and a starting price.
 - Bidders can place bids on items.
 - Admin monitors and finalizes the winner based on the highest bid.
 - Real-time bidding updates via WebSocket.

3.4.5 Report Module

- **Features:**

Generate reports for:

- Past auctions, including details of bids and winners.
- Sellers and their product listings.
- Bidders and their activity.

3.5 User Specifications

3.5.1 Admin

- **Responsibilities:**
 - Manage users (Sellers and Bidders) and their accounts.
 - Manage categories for organizing products.
 - Oversee auctions and ensure fair bidding.
 - Manage packages available for bidders.
 - Send updates and notifications to sellers and bidders.
 - Generate system-wide reports, including auction outcomes, user activities, and overall performance.

3.5.2 Seller

- **Responsibilities:**
 - Upload products for auction with necessary details (title, description, images, starting price,time).
 - Set the starting price for their products.
 - View bid information and activity for their listed items.
 - Participate as a bidder for other auctions if needed.

3.5.3 Bidder

- **Responsibilities:**
 - Search for auction items using category or keyword filters.
 - Purchase packages to gain bidding access.
 - View detailed product information, including descriptions and current bids.
 - Place bids on products and modify their bid price if necessary.
 - Monitor auction status and receive notifications about bidding updates.

3.6 Non-Functional Requirements

3.6.1 Security

- Secure user authentication and authorization using Firebase Authentication.
- Data encryption for sensitive information (e.g., passwords, bids, and transactions).

3.6.2 Reliability

- High system uptime to ensure continuous availability.
- Robust error handling to maintain functionality during unexpected events.

3.6.3 Performance

- Real-time bidding updates with low latency using WebSocket.
- Quick search and filter operations for products and categories.

3.6.4 Usability

- Intuitive and user-friendly interface for all user types.
- Seamless navigation and clear instructions for performing tasks.

3.6.5 Scalability

- Scalable database design (Firebase Firestore) to handle increasing users, items, and bids.

3.6.6 Compatibility

- Mobile compatibility for seamless access on smartphones.
- Support for modern browsers for web-based components.

3.7 Stakeholder Expectations

3.7.1 Admin:

- Easy-to-use admin dashboard for managing users, products, categories, and auctions.
- Ability to send updates to users and generate detailed reports.

3.7.2 Seller:

- Seamless product listing experience with real-time visibility of bids.
- Flexibility to participate as a bidder if desired.

3.7.3 Bidder:

- Smooth bidding experience with instant updates on the auction status.
- Detailed product information and package purchasing options.

3.8 Use Case Diagram

The **Use Case Diagram** illustrates the interactions between the system and its users. For the **BidSell platform**, it highlights two primary panels:

1. **User Panel:** Shows how buyers and sellers interact with the platform. Buyers can search, view auction details, place bids, and manage their profiles. Sellers can create auctions and track bidding activities.
2. **Admin Panel:** Depicts the admin's responsibilities, such as managing users, approving auctions, overseeing transactions, and sending notifications to maintain a secure and transparent platform.

3.8.1 Use Case Diagram for User panel

This is the use case diagram for our users, showcasing the various tasks they can perform within the system.

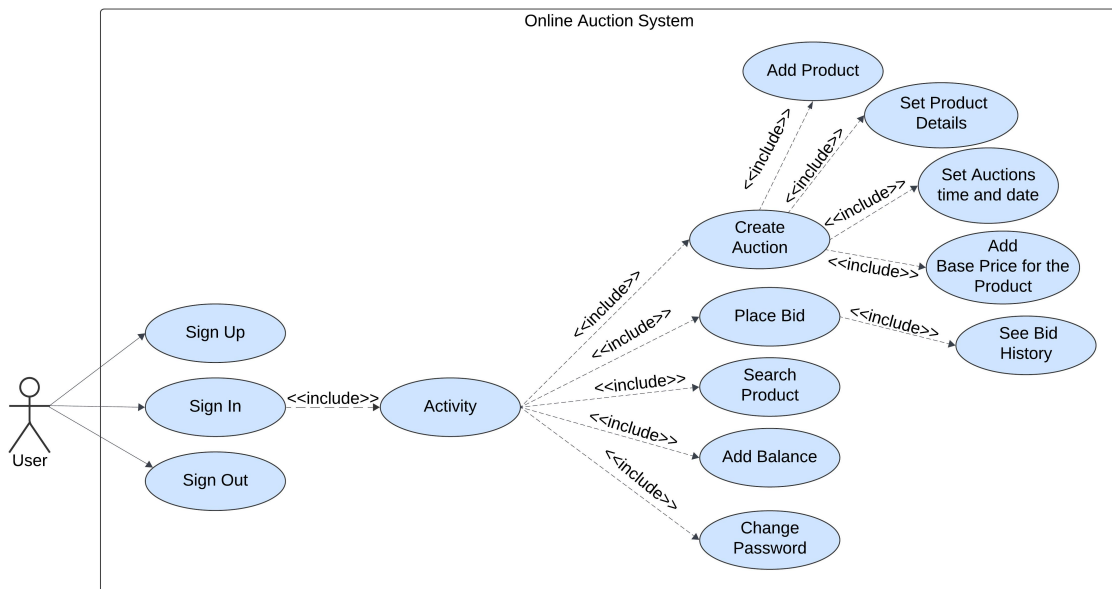


Figure 3.2: Use Case Diagram for User panel

3.8.1.1. Actor:

- **User:** Represents the person interacting with the online auction system.

3.8.1.2. System Boundary:

- The rectangle labeled **Online Auction System** defines the boundary of the system, showing what functionalities are included.

3.8.1.3. Use Cases:

- Represented by oval shapes, these depict the functions or actions users can perform in the system. Key use cases include:
 - **Sign Up:** Allows a user to register for the system.
 - **Sign In:** Enables a registered user to log into the system.
 - **Sign Out:** Allows a user to log out of the system.
 - **Activity:** Represents a collection of primary actions the user can perform after signing in.
 - **Create Auction:** Users can set up an auction for a product.
 - **Place Bid:** Users can place a bid on available products.
 - **Search Product:** Enables users to search for specific products in the auction.
 - **Add Balance:** Users can add funds to their account for bidding.
 - **Change Password:** Users can update their login credentials.

3.8.1.4. Sub-functions with <<include>>:

The **include** relationship shows that some use cases depend on other, smaller tasks to be completed. For example:

- **Create Auction:**
 - Includes sub-tasks like **Add Product**, **Set Product Details**, **Set Auctions Time and Date**, and **Add Base Price for the Product**.
- **Place Bid:**
 - Includes **See Bid History**, allowing users to view previous bids before placing a new one.

3.8.1.5. Relationships:

- **Solid Lines:** Represent direct interaction between the actor and the use case.
- **Dashed Arrows with <<include>>:** Show mandatory dependencies, meaning the base use case cannot function without the included use case. For example:
 - To "Create Auction," it is mandatory to "Add Product" and "Set Auction Details."

3.8.2 Use Case Diagram for Admin Panel

This is the use case diagram for our users, showcasing the various tasks they can perform within the system.

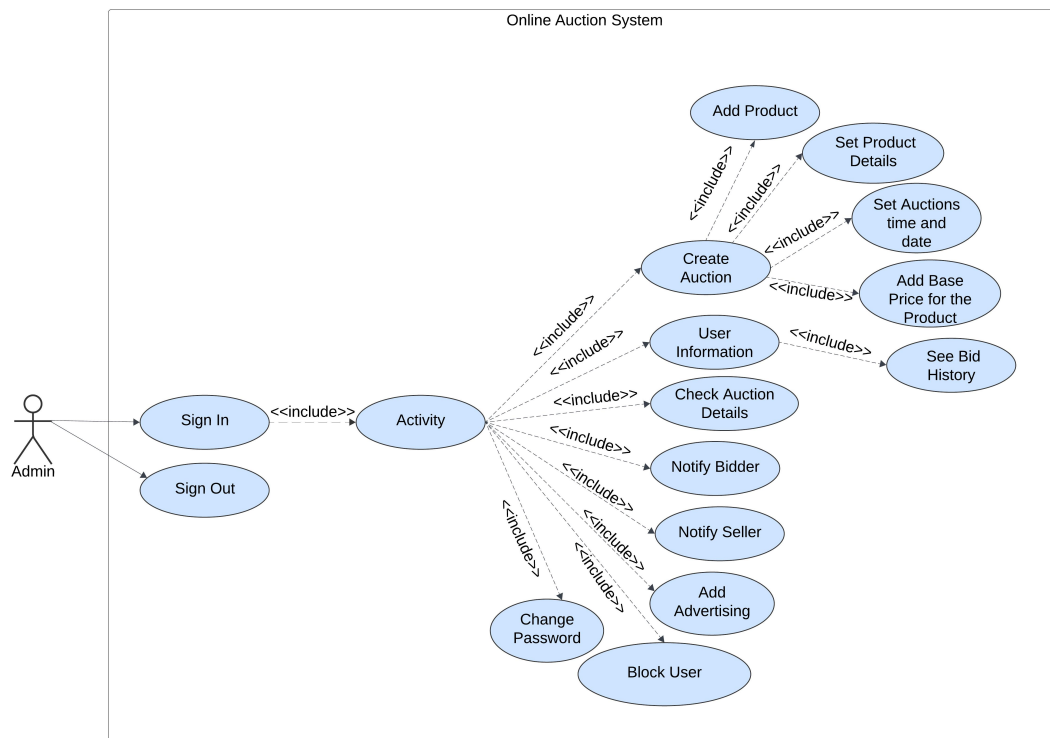


Figure 3.3: Use Case Diagram for Admin panel

3.8.2.1 Actors:

- **Admin:** The **Admin** is responsible for managing the entire system.

3.8.2.2 Use Cases:

The system supports various functionalities for the Admin, categorized as follows:

3.8.2.2.1 Account Management:

- **Sign In:** Admin logs into the system to access its functionalities.
- **Sign Out:** Admin logs out after completing their tasks.
- **Change Password:** Admin can update their login credentials for security purposes.

3.8.2.2.2 Auction Management:

- **Create Auction:** The Admin sets up new auctions in the system.
 - **Set Product Details:** Define the attributes of a product (e.g., name, description, and category).
 - **Set Auction Time and Date:** Schedule the start and end times for the auction.
 - **Add Base Price for the Product:** Specify the minimum bidding price for the product.
- **Check Auction Details:** Review the details of ongoing or completed auctions.
- **See Bid History:** View the history of bids placed on products.
- **Notify Bidder:** Send notifications to bidders regarding auction updates.
- **Notify Seller:** Send notifications to sellers about their products or bids.

3.8.2.2.3 Product Management:

- **Add Product:** Admin can add new products to the auction.
- **Add Advertising:** Promote products or auctions through advertising within the platform.

3.8.2.2.4 User Management:

- **User Information:** Access user profiles and details.
- **Block User:** Restrict or ban users who violate platform rules.

3.9 Data Flow Diagram (DFD)

A **Data Flow Diagram (DFD)** represents the flow of data within a system, showcasing how inputs are processed and outputs are generated. It illustrates the relationships between different components, including:

- **Processes:** Actions or functions performed on the data.
- **Data Stores:** Locations where data is stored.
- **External Entities:** Sources or destinations of data outside the system.

- **Data Flows:** Movement of data between entities, processes, and data stores.

DFDs provide a clear and simplified view of the system's operations, helping to analyze, design, and understand data movement and processing.

3.9.1 DFD level 0

This is the Level 0 Data Flow Diagram, representing the overall system and its interactions with external entities.

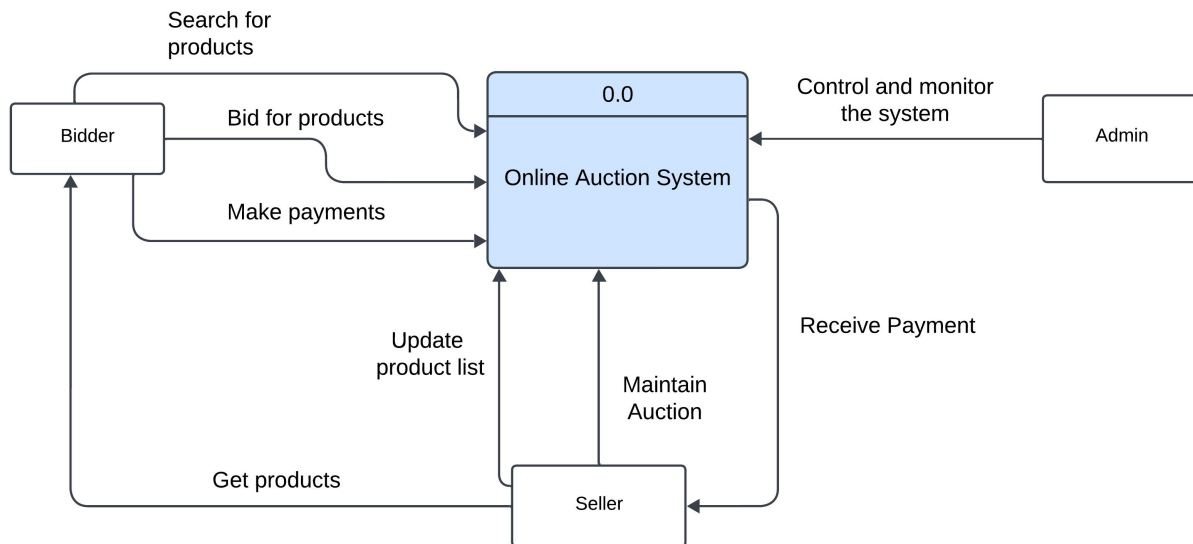


Figure 3.4: DFD level 0

The **Level 0 Data Flow Diagram (DFD)** provides a high-level overview of the system's data flow. It depicts the interactions between key external entities, such as:

- **Bidder:** Searches for products, places bids, and makes payments.
- **Seller:** Updates product listings, provides products, and maintains auction start/end times.
- **Admin:** Monitors and controls the system. The central system processes these interactions, ensuring smooth operation and data exchange between entities.

3.9.2 DFD level 1

This is the Level 1 Data Flow Diagram, breaking down the system processes into detailed sub-processes and their interactions with external entities.

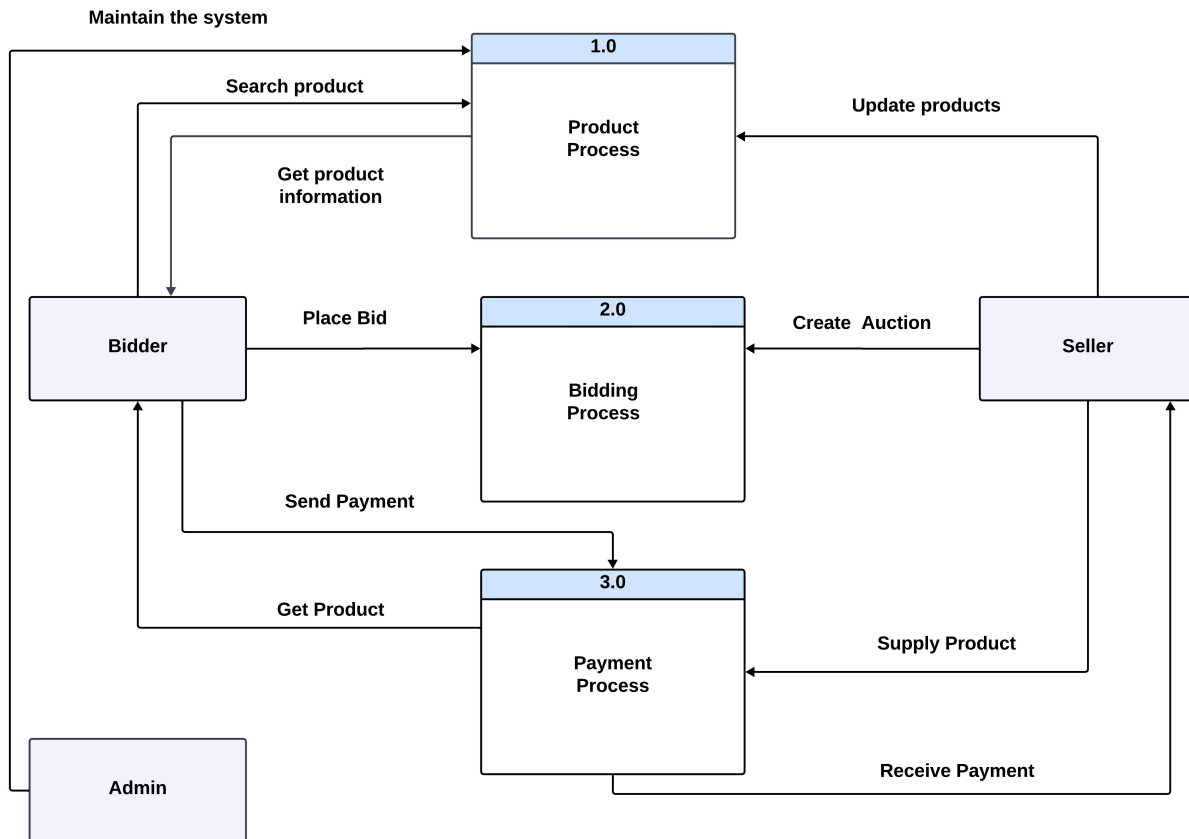


Figure 3.5: DFD level 1

The **Level 1 Data Flow Diagram (DFD)** provides a detailed view of the system's processes and their interactions with external entities. Key components include:

- **Processes:**
 - **1.0 Product Process:** Handles product search, product information retrieval, and updates from sellers.
 - **2.0 Bidding Process:** Manages bids placed by bidders and communicates with sellers for auction updates.
 - **3.0 Payment Process:** Processes payments from bidders and ensures sellers receive payment after product delivery.
- **External Entities:**
 - **Bidder:** Searches products, makes bids, sends payments, and receives purchased products.
 - **Seller:** Updates product details, manages auction information, and supplies products.
 - **Admin:** Monitors and maintains the system.

- **Data Flows:**
 - **Search Product:** From bidder to product process.
 - **Make a Bid:** From bidder to bidding process.
 - **Send Payment:** From bidder to payment process.
 - **Supply Product:** From seller to payment process.
 - **Update Products:** From seller to product process.

3.10 Entity Relationship Diagram (ER Diagram)

This is the Entity-Relationship (ER) Diagram, representing the database structure and relationships between entities in the system.

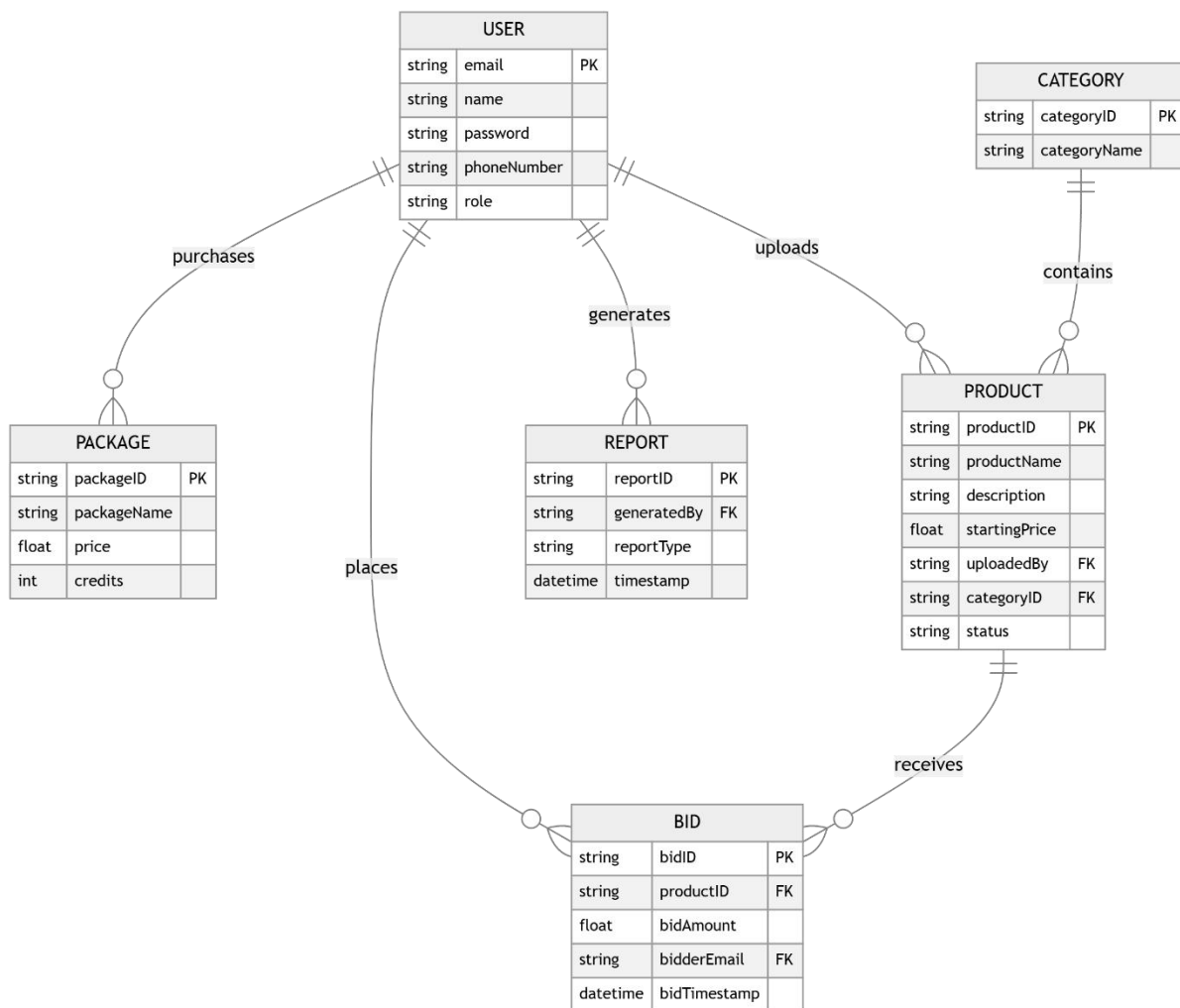


Figure 3.6: ER Diagram

The **Entity-Relationship (ER) Diagram** outlines the database structure, detailing entities and their relationships:

- **Entities:**

- **USER:** Stores user details like email (PK), name, password, phone number, and role.
- **PACKAGE:** Contains information about packages purchased by users, including packageID (PK), name, price, and credits.
- **PRODUCT:** Represents products uploaded by users, with attributes such as productID (PK), name, description, starting price, uploadedBy (FK), categoryID (FK), and status.
- **CATEGORY:** Categorizes products using categoryID (PK) and categoryName.
- **REPORT:** Logs reports generated by users, including reportID (PK), generatedBy (FK), type, and timestamp.
- **BID:** Tracks bids placed on products, with bidID (PK), productID (FK), bidAmount, bidderEmail (FK), and timestamp.

- **Relationships:**

- **USER purchases PACKAGE:** Tracks the packages purchased by users.
- **USER uploads PRODUCT:** Represents products uploaded by users.
- **PRODUCT contains CATEGORY:** Associates products with specific categories.
- **USER generates REPORT:** Logs reports created by users.
- **USER places BID:** Tracks bids placed by users on products.
- **PRODUCT receives BID:** Links bids to specific products.

3.11 Sequence Diagram

A Sequence Diagram is a type of UML (Unified Modeling Language) diagram that visualizes the sequence of interactions or messages between objects in a system to perform a specific task.

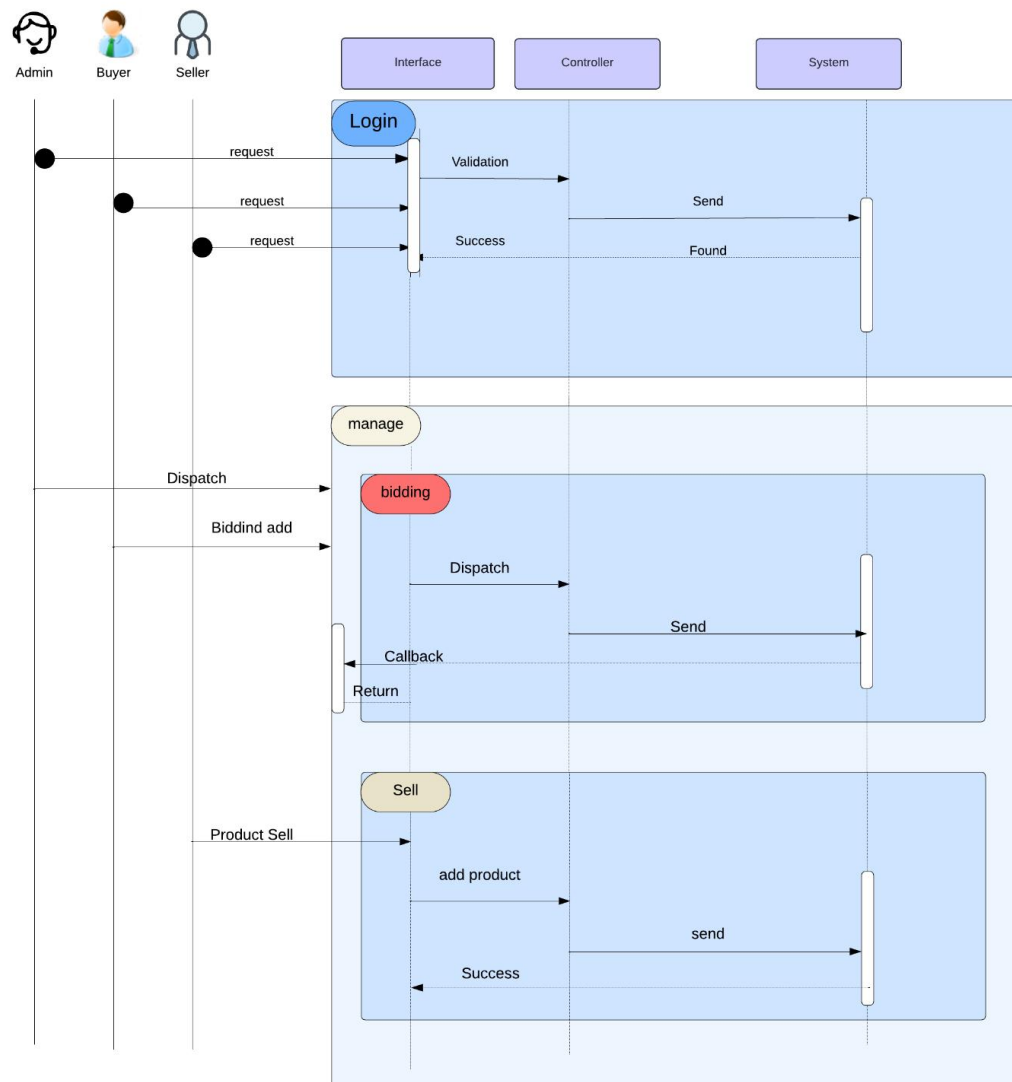


Figure 3.7: Sequence Diagram

The **Sequence Diagram** illustrates the flow of requests and responses between the system's actors, namely Admin, Buyer, and Seller, and the system's components, including the Interface, Controller, and System. The login process begins with all users (Admin, Buyer, Seller) sending a login request, which is validated by the system. Upon successful validation, the system grants access to the user. In the bidding process, buyers manage their bids by dispatching a request. The system processes this request and provides a response, ensuring callback and successful bid placement. For product selling, sellers add new products to the system by sending a product request. The system processes this request, adds the product, and confirms the success of the operation. This sequence diagram provides a comprehensive view of how key functionalities are handled within the system.

3.12 System Flow Chart

This system flowchart represents the step-by-step flow of user actions (Admin, Bidder, and Seller) and their interactions within the platform.

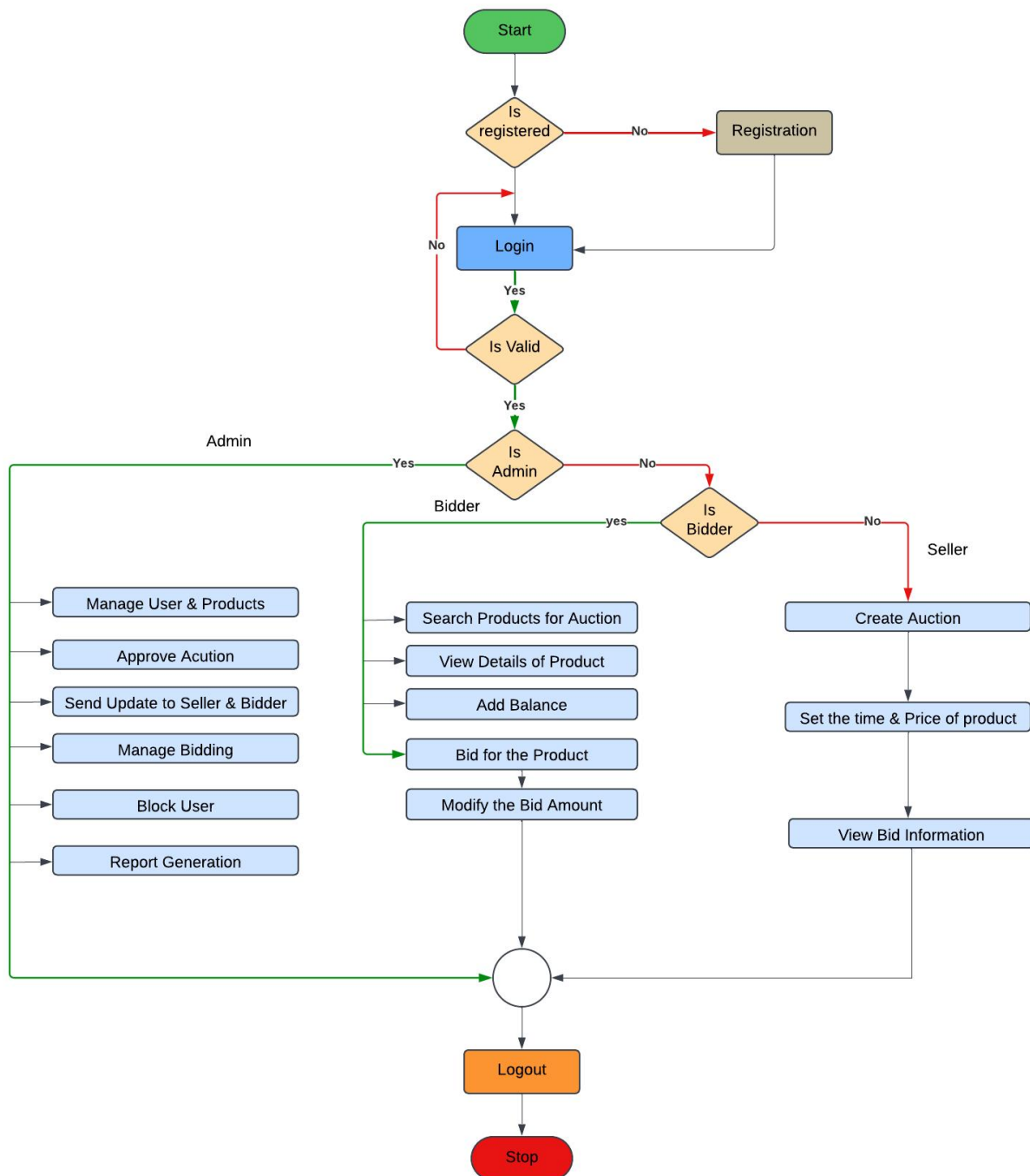


Figure 3.8: System Flow Chart

The **System Flowchart** begins with the user determining their registration status. If not registered, they are redirected to the registration process. Upon successful login and validation, the system identifies the user role (Admin, Bidder, or Seller) and provides specific functionalities:

- **Admin:** Admin users can manage users and auctions, approve auctions, send updates to sellers and bidders, manage bidding activities, block users, and generate reports.
- **Bidder:** Bidders can search for auction products, view product details, add balance to their accounts, place bids on products, and modify bid amounts.
- **Seller:** Sellers can create auctions by providing necessary details, such as auction time, date, and base price, and view bid information for their products.

The flow concludes with a logout process, ensuring a secure exit from the system. This chart provides a clear visualization of the operational workflow for all user roles.

3.13 Conclusion

We outlined various software development methodologies and their applicability to the auction app. While methodologies like Waterfall and V-Shaped models offer structured approaches, they lack the flexibility required for this dynamic project. Incremental, Iterative, Spiral, and RAD models provide iterative improvements but are less suited compared to Agile. The Agile methodology emerged as the ideal choice for the auction app due to its adaptability, user-centric focus, and iterative delivery approach. It allows the project to respond effectively to evolving requirements, incorporate user feedback, and deliver high-priority features incrementally. Agile ensures a collaborative and efficient development process, aligning with the app's need for real-time updates, secure transactions, and scalability. Additionally, functional and non-functional requirements, including modules for login, auction management, and real-time bidding, were defined, ensuring a clear roadmap for implementation. Use case, data flow, and system diagrams were presented to visualize the app's design and workflows comprehensively. Together, these elements establish a strong foundation for the development of a robust and user-friendly auction platform.

Chapter 4

Implementation

4.1 Introduction

The implementation phase translates the planned methodologies and designs into a functional auction application. This stage focuses on bringing the defined features and requirements to life, using the Agile methodology for iterative development and user-centric design. Key modules such as user authentication, auction management, real-time bidding, and payment integration are built and integrated systematically. The process involves leveraging modern technologies, tools, and frameworks to ensure scalability, security, and performance. Close collaboration among developers, designers, and stakeholders drives continuous improvements, aligning the application with user needs and expectations. This phase is critical in turning the conceptual design into a practical, robust, and efficient system for facilitating seamless auctions.

4.2 Features for Users panel

The **User Panel** in the BidSell application provides buyers and sellers with essential tools for seamless interaction:

- **For Buyers:** Users can browse auctions, place bids in real time, view auction histories, and manage profiles. Notifications ensure they stay updated on bid changes.
- **For Sellers:** Allows uploading items for auction, managing product listings, and tracking bidding activity on their items.
- **Additional Features:** Secure multi-option authentication, password recovery via email/OTP, intuitive user interface, profile management, mobile compatibility, and multi-layered security for enhanced user experience.

4.2.1 OnBoarding Activity

This screen serves as the gateway to BidSell, introducing users to the platform with options to sign up or sign in. It reflects the brand identity and provides a seamless starting point for users to access the online auction system.

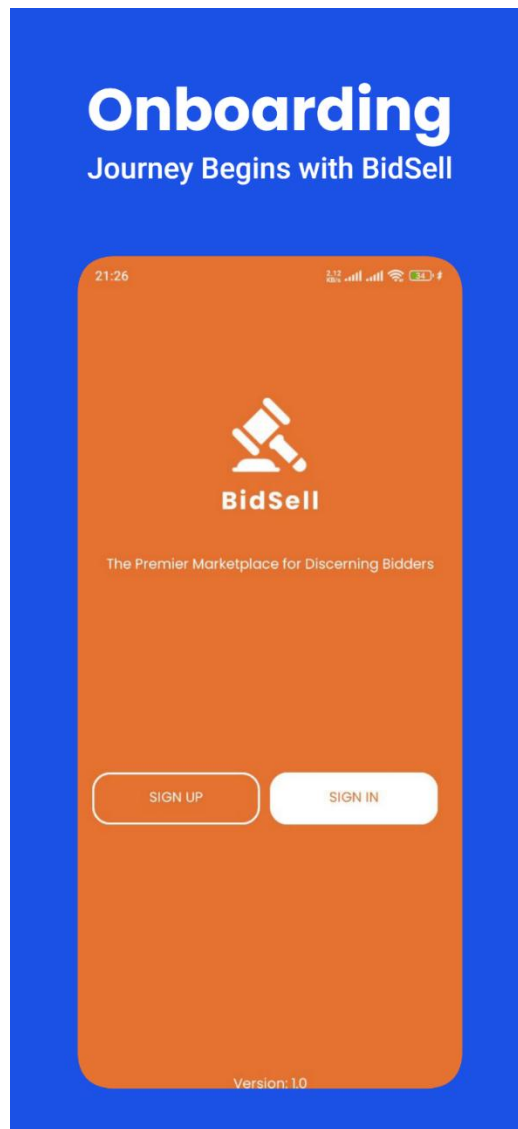


Figure 4.1: OnBoarding Activity

This onboarding screen welcomes users to the BidSell platform, allowing new users to register or existing users to log in. It provides quick access to start bidding or managing auctions seamlessly.

4.2.2 Sign Up Activity

This is the registration screen of the BidSell platform. It collects user details to create a secure account for platform access.

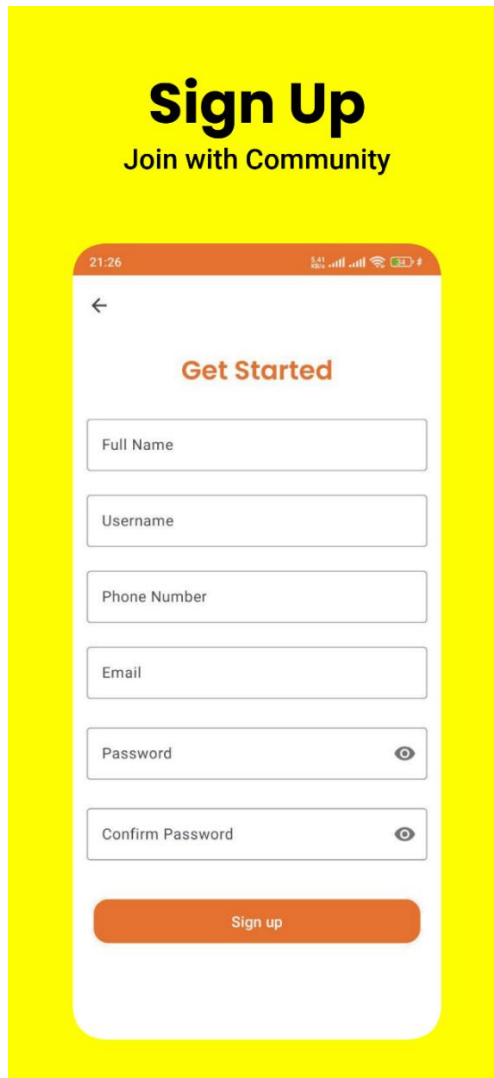


Figure 4.2: Sign Up Activity

The screen represents the registration page of the BidSell application. It enables new users to join the platform by filling out fields such as Full Name, Username, Phone Number, Email, Password, and Confirm Password. This information is used to create a secure user account, ensuring safe access and personalized experiences. The screen is designed to be intuitive and user-friendly, streamlining the onboarding process.

4.2.3 Sign In Activity

This is the sign-in screen of the BidSell app. Existing users can log in by entering their email and password to access the platform.

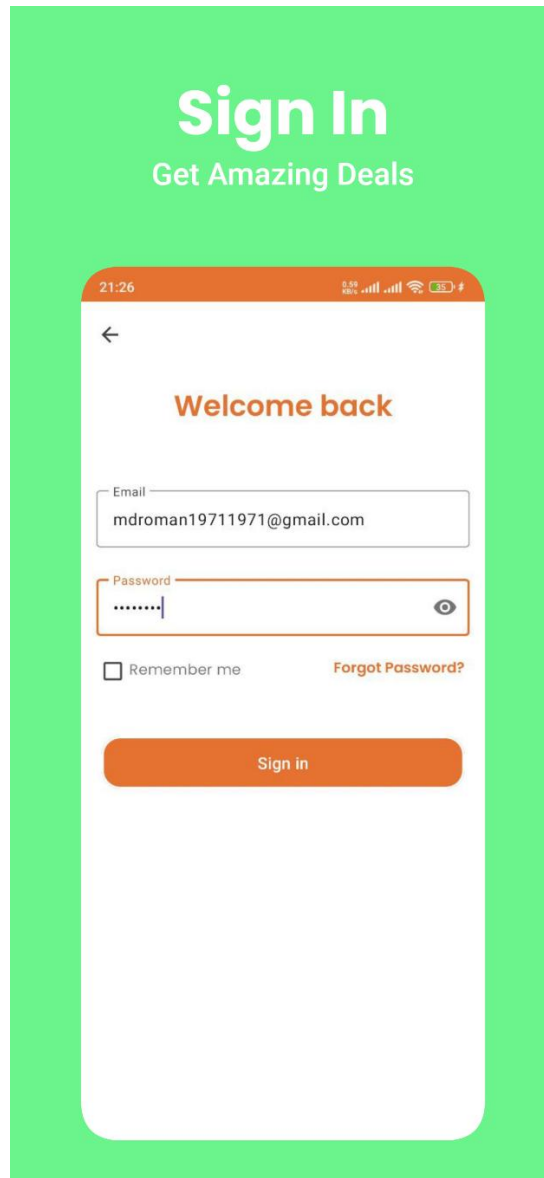


Figure 4.3: Sign In Activity

The screen represents the sign-in page of the BidSell application. Users can log in by providing their email and password. It features a "Remember me" option for convenience and a "Forgot Password?" link for users who need to recover their password. This login screen is designed to ensure secure and easy access to the app for returning users, allowing them to quickly return to bidding or managing their auctions.

4.2.4 Home Screen Activity

This is the auction discovery screen of the BidSell app. It allows users to explore upcoming, running, and closed auctions effortlessly.

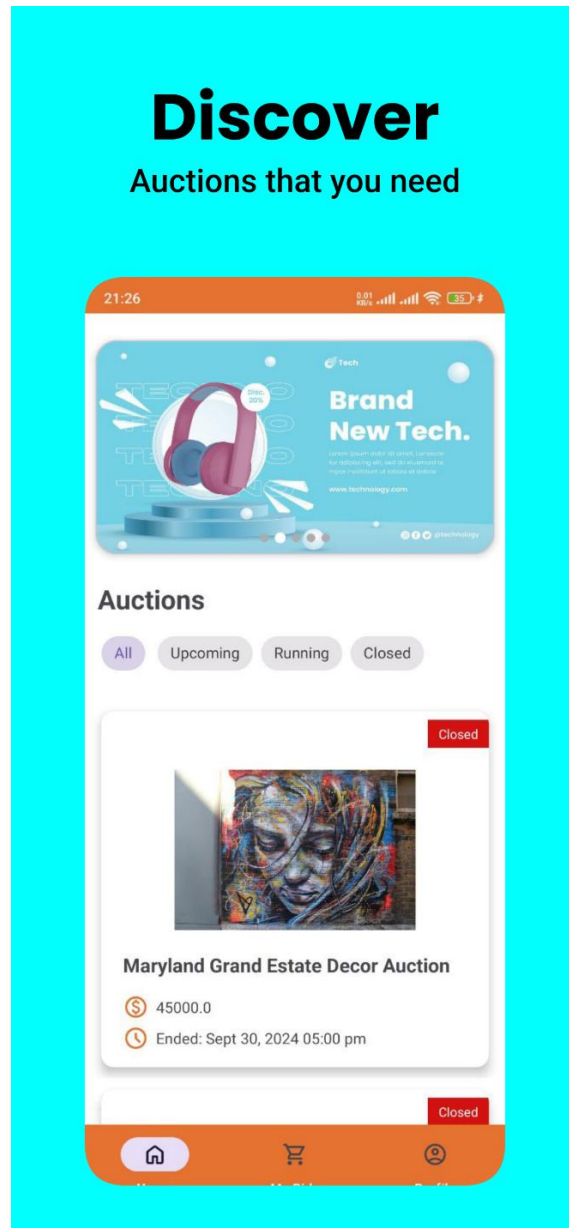


Figure 4.4: Home Screen Activity

The screen showcases the auction discovery page of the BidSell application. It provides a clean and organized layout, displaying auctions categorized as All, Upcoming, Running, or Closed. Each auction is presented with an image, price details, and its status, ensuring users can quickly access the information they need. This design ensures a seamless browsing experience and helps users navigate the auction options with ease.

4.2.5 Auction Details Activity

This is the auction details screen of the BidSell app. It highlights featured auctions with detailed information and history.

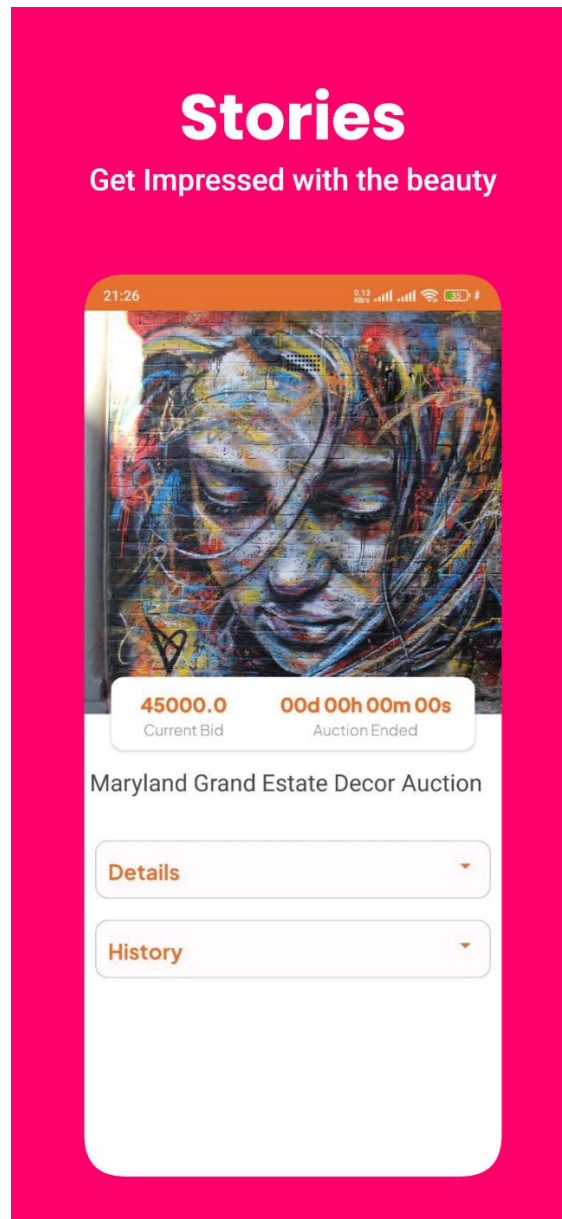


Figure 4.5: Auction Details Activity

The screen represents the auction details page of the BidSell application, focusing on a visually engaging display. It showcases the featured auction with high-quality visuals, the current bid amount, and the auction status. Users can access additional details and the bidding history through expandable sections, enhancing user engagement and providing all relevant information in an intuitive layout. This design ensures clarity and encourages active participation in the auction process.

4.2.6 All Auctions Activity

This is the ongoing auctions screen of the BidSell app. Users can browse active auctions and access key information at a glance.

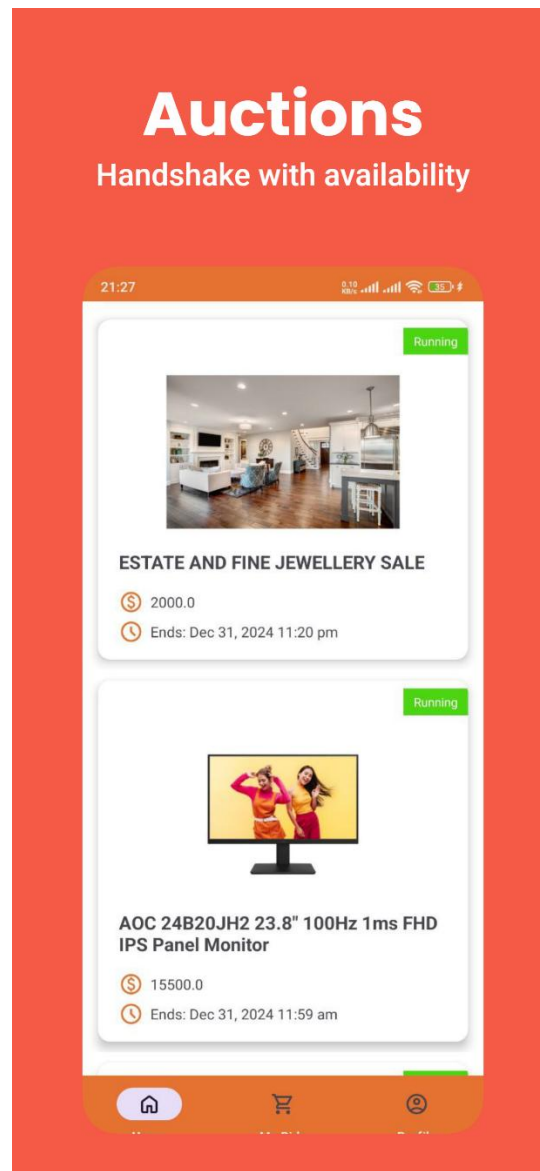


Figure 4.6: All Auctions Activity

The screen showcases the ongoing auctions page of the BidSell application, listing current auctions with essential details such as item descriptions, current bid amounts, and auction end times. Each auction is marked with its status, like "Running," for clarity. The intuitive layout ensures easy navigation, allowing users to quickly explore and participate in live auctions, making the bidding process seamless and user-friendly.

4.2.7 History Module

This is the history screen of the BidSell app, allowing users to review detailed auction and bid records.

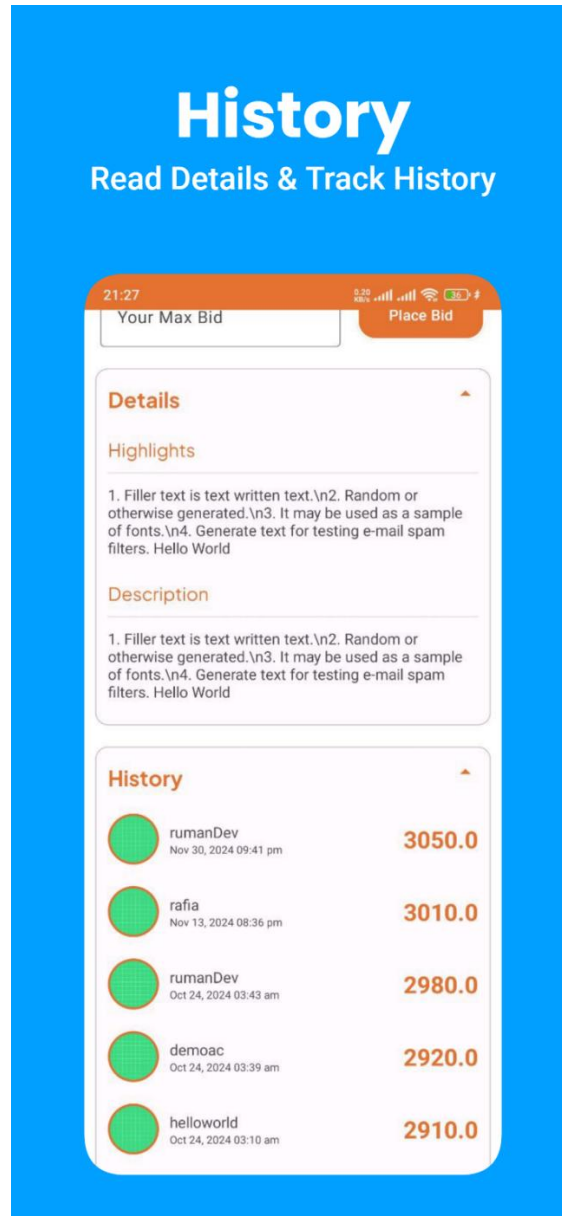


Figure 4.7: History Module

The screen displays the auction history page of the BidSell application, providing users with a comprehensive view of past activities. It includes sections for auction details, highlights, and descriptions, along with a history log that lists bid amounts, usernames, and timestamps. The design ensures clarity and ease of use, allowing users to track their bidding history and analyze previous auction outcomes efficiently. This feature enhances transparency and helps users make informed decisions.

4.2.8 User Profile Activity

This is the features screen of the BidSell app, providing users access to essential app functionalities in one place.

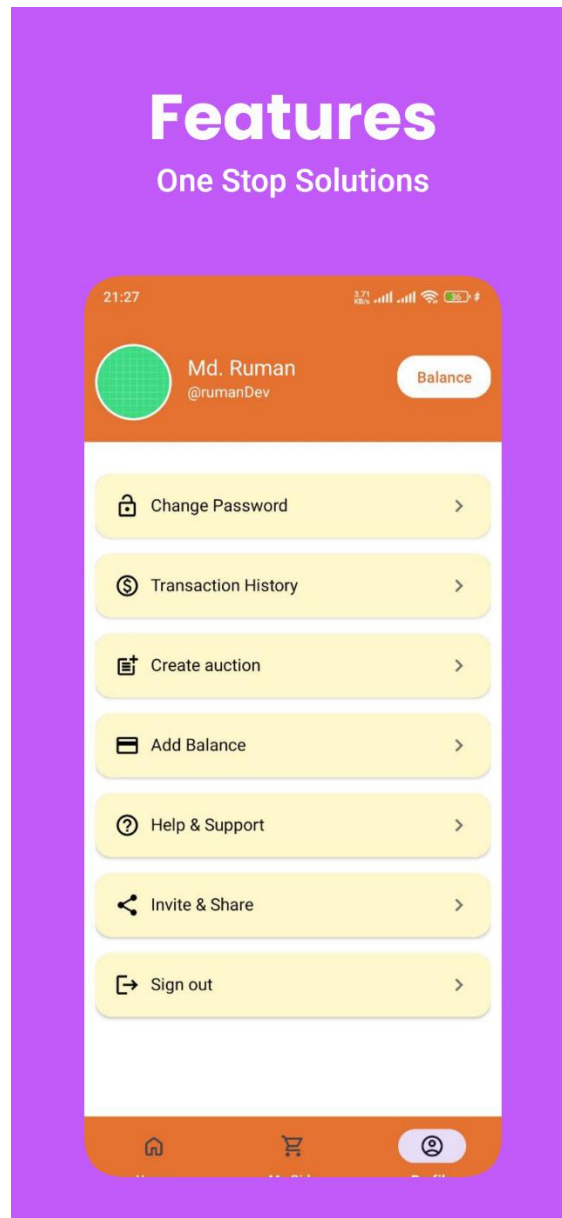


Figure 4.8: User Profile Activity

The screen displays the profile and features page of the BidSell application, offering key functionalities such as changing passwords, viewing transaction history, creating auctions, adding balance, and accessing help and support. It also includes options to invite and share the app and to sign out. The simple and intuitive menu design ensures seamless navigation, enabling users to manage their accounts and activities efficiently. This centralized functionality enhances the overall user experience.

4.2.9 Change Password Module

This is the password change screen of the BidSell app, designed to ensure user security and privacy.

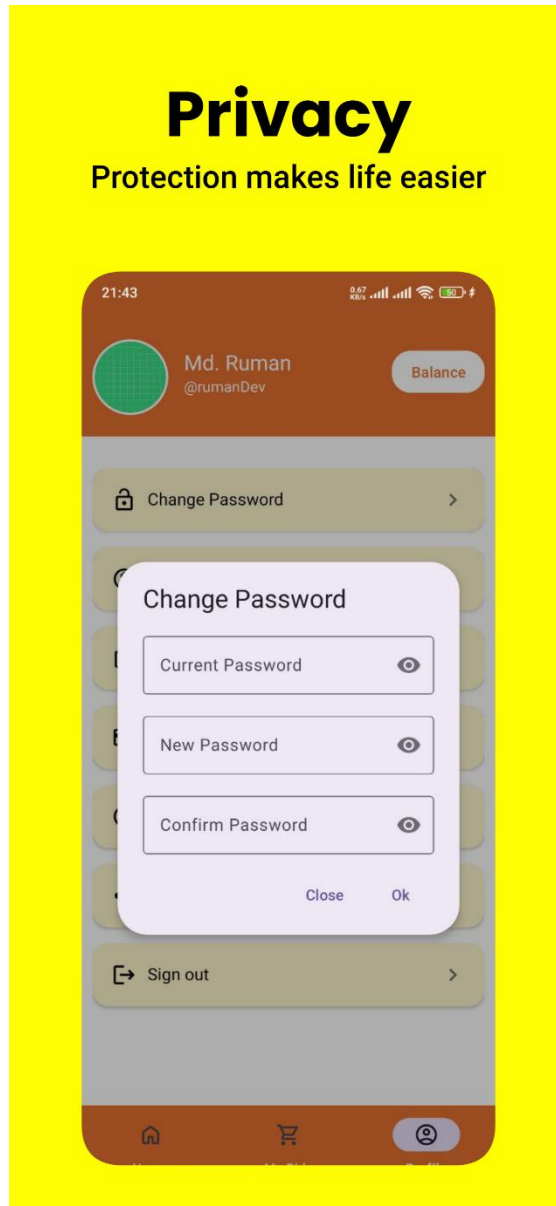


Figure 4.9: Change Password Module

The screen showcases the "Change Password" feature of the BidSell application, allowing users to update their account credentials securely. It includes input fields for the current password, a new password, and confirmation of the new password. The straightforward interface prioritizes security and ease of use, ensuring users can manage their account settings effortlessly and protect their personal information.

4.2.10 Request Auction Activity

This is the auction creation screen of the BidSell app, designed to facilitate seamless auction setup by users.

Earn
Rewards by Creating Auction

21:27 5.72 26% 5G

Create Auction Form

Auction Name

Starting Price

Image Link
* Image format must be enabled at last such as jpg, png

Highlights Text

Description Text

Quantity

Opening Time

Closing Time

Submit a request

Figure 4.10: Request Auction Activity

The screen displays the "Create Auction Form" of the BidSell application, enabling users to set up auctions with ease. It includes input fields for Auction Name, Starting Price, Image Link, Highlights, Description, Quantity, Opening Time, and Closing Time. The minimalistic design ensures efficient data entry and submission, making the process user-friendly and straightforward. This feature empowers users to list their items for bidding quickly and effectively.

4.2.11 Share App

This is the sharing screen of the BidSell app, allowing users to invite friends and expand the community.

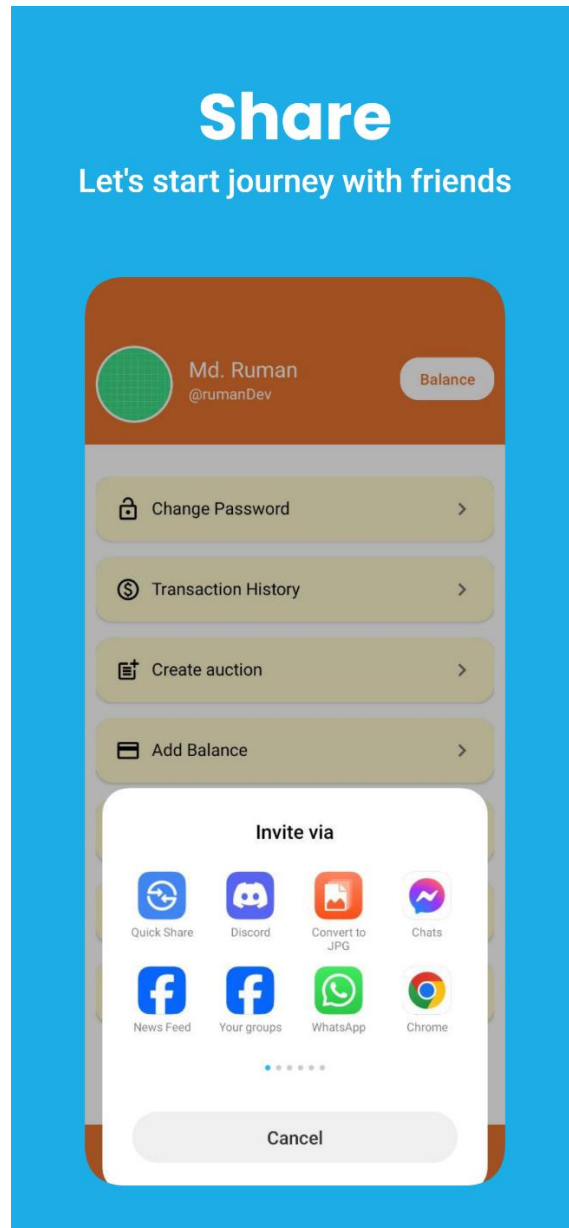
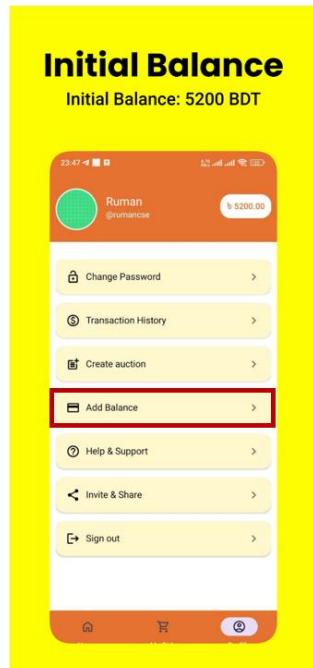


Figure 4.11: Share App Activity

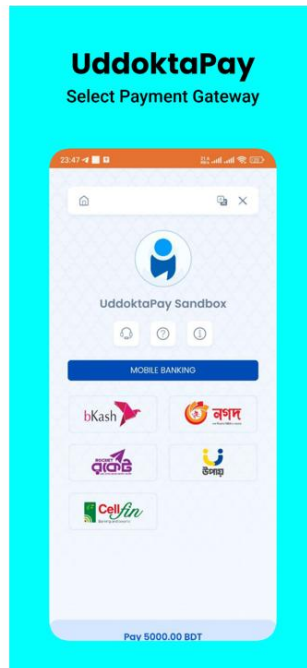
The screen showcases the sharing feature of the BidSell application, providing users with multiple options to share the app or auction details. Users can invite friends via platforms like Discord, WhatsApp, Facebook, and more, directly from this screen. The design is intuitive and ensures easy navigation, encouraging users to expand the app's reach by inviting others to join the BidSell platform. This feature fosters community growth and engagement.

4.2.12 Add Balance Module

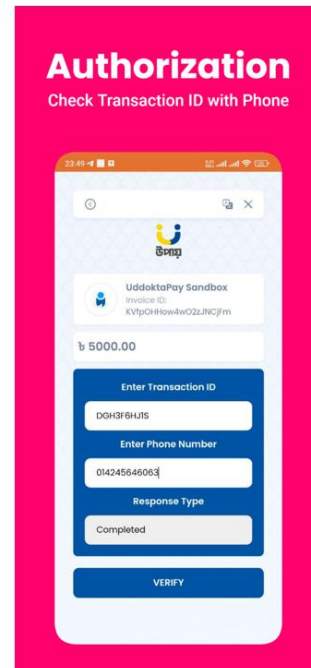
The Add Balance module of the BidSell app allows users to seamlessly add funds to their accounts. Below are the steps demonstrated in the images:



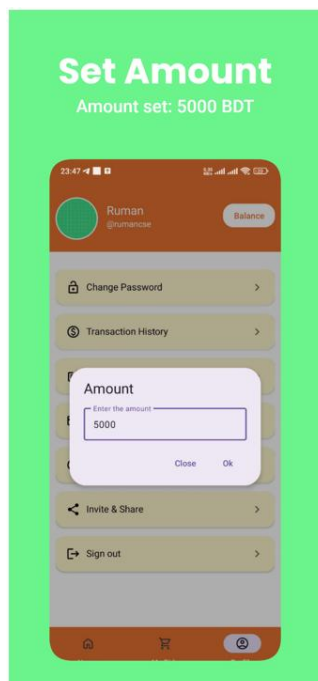
Step - 01



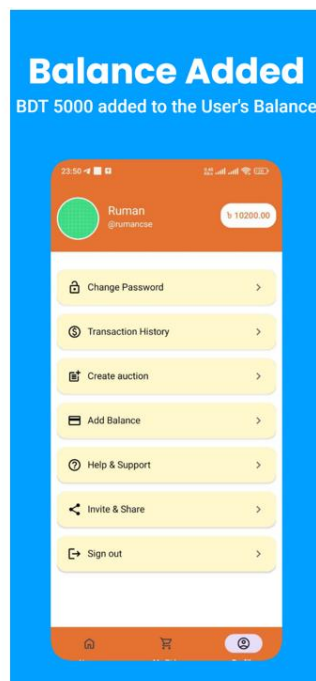
Step - 02



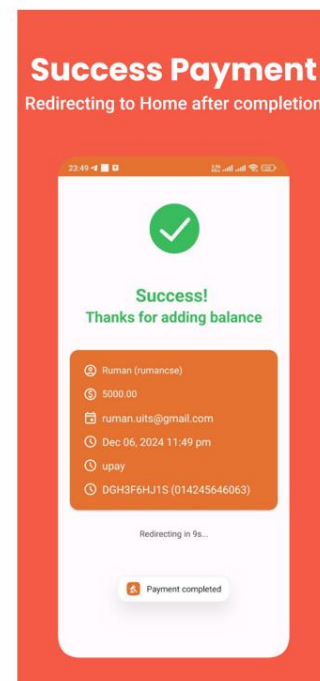
Step - 03



Step - 04



Step - 05



Step - 06

Figure 4.12: Add Balance Module

Step 01: Initial Balance

The user begins with an overview of their current balance on the profile page. In this example, the initial balance is **5200 BDT**. By selecting the "Add Balance" option, users can initiate the process of adding funds.

Step 02: UddoktaPay Payment Gateway

Users are redirected to the UddoktaPay payment gateway, where they can choose their preferred payment method, such as mobile banking (e.g., **bKash**, **Nagad**, or **Rocket**). This ensures a variety of options for seamless transactions.

Step 03: Authorization

The user enters the transaction ID and phone number for verification. Once verified, the payment gateway processes the transaction securely. This step ensures that the transaction is both validated and secure.

Step 04: Set Amount

The user specifies the amount they want to add, which in this example is **5000 BDT**. This amount is displayed in a confirmation form for review, allowing the user to verify before proceeding.

Step 05: Balance Added

After a successful transaction, the specified amount (**5000 BDT**) is added to the user's account. The updated balance is now **10200 BDT** (5200 BDT initial balance + 5000 BDT added). This updated balance is immediately reflected on the profile page.

Step 06: Success Payment

The user receives a success message summarizing the transaction details, including:

- The amount added (**5000 BDT**).
- Updated balance (**10200 BDT**).
- Date, time, and payment method used (e.g., Upay).
- Transaction ID.

4.3 Features for Admin Panel

The **Admin Panel** of the BidSell app provides comprehensive control over the platform, ensuring smooth operations and user management. Key features include:

- **User Management:** Manage buyer and seller accounts, approve registrations, and block violators.
- **Auction Management:** Approve or reject auctions, update details, and monitor activities.
- **Bid Management:** Review bids, verify authenticity, and resolve disputes.
- **Transaction Management:** Track payments, view histories, and generate financial reports.
- **Category Management:** Organize items into categories for easy navigation.
- **Notify Management:** Send alerts, updates, and reminders to users.
- **Advertising Management:** Manage sponsored content and generate revenue.
- **Reports and Analytics:** Generate insights into platform performance and user activities.
- **Help and Support:** Address user queries and feedback efficiently.
- **System Settings:** Manage platform policies, permissions, and technical configurations.

4.3.1 Admin Login Activity

This is the login screen for the BidSell Admin Panel, allowing administrators secure access to manage the platform.

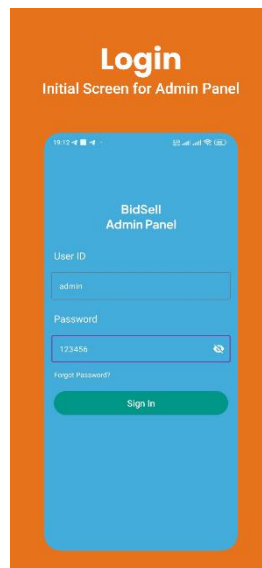


Figure 4.13: Admin Login Activity

The screen displays the login interface for the BidSell Admin Panel. Administrators can log in using their credentials, such as User ID and Password, to access platform management tools. The interface includes features like a "Forgot Password?" link for account recovery and a password visibility toggle for ease of use. This screen ensures secure and streamlined access for platform administrators.

4.3.2 Admin Dashboard Activity

This is the Admin Dashboard screen of the BidSell app, providing an overview of platform activity and statistics.

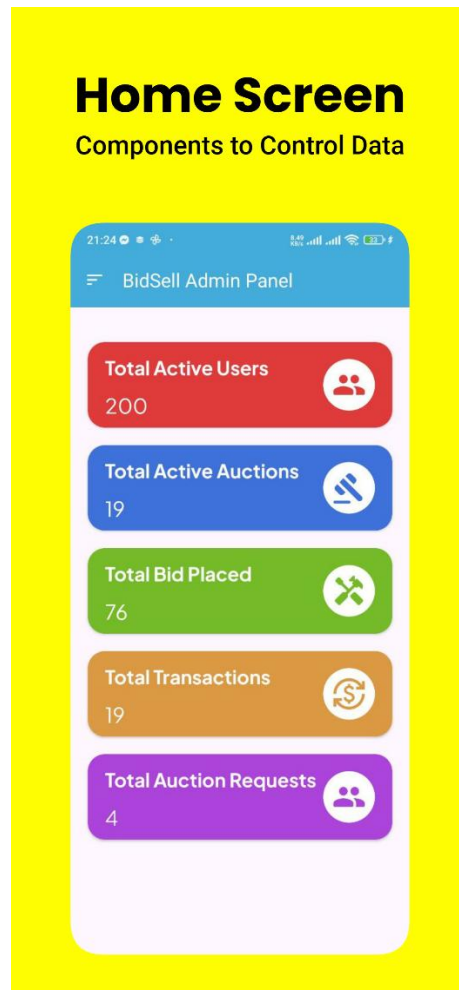


Figure 4.14: Admin Dashboard Activity

The **Admin Dashboard** provides a comprehensive overview of the platform's key statistics for efficient management. It includes the following highlights:

- **Total Active Users:** Displays the number of users currently registered on the platform.
- **Total Active Auctions:** Shows the total number of ongoing auctions.
- **Total Bids Placed:** Summarizes the bidding activity across all auctions.
- **Total Transactions:** Tracks the number of completed financial transactions.
- **Total Auction Requests:** Lists the pending auction requests awaiting approval.

4.3.3 User Info Activity

This is the Manage Users screen of the BidSell Admin Panel, offering centralized control over user accounts.

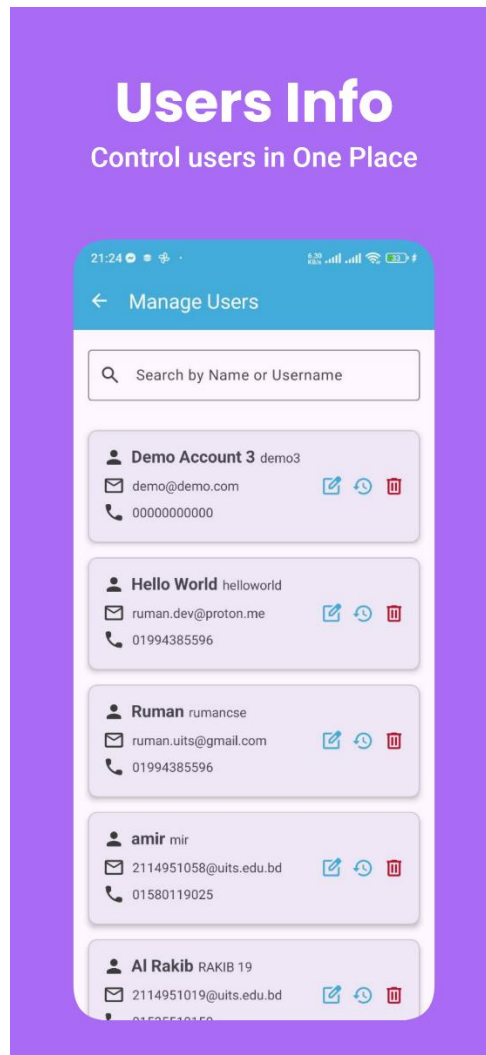


Figure 4.15: User Info Activity

The Manage Users screen allows administrators to view and manage user accounts in one place. Key functionalities include:

- **Search by Name or Username:** Quickly locate specific user accounts.
- **Edit User Information:** Update user details such as email or phone number.
- **User Communicate:** Communicate directly with users
- **Delete Accounts:** Remove accounts violating platform policies or inactive users.

4.3.4 Manage Auction Request Activity

This is the Auction Request screen of the BidSell Admin Panel, designed for reviewing and managing auction submissions from users.

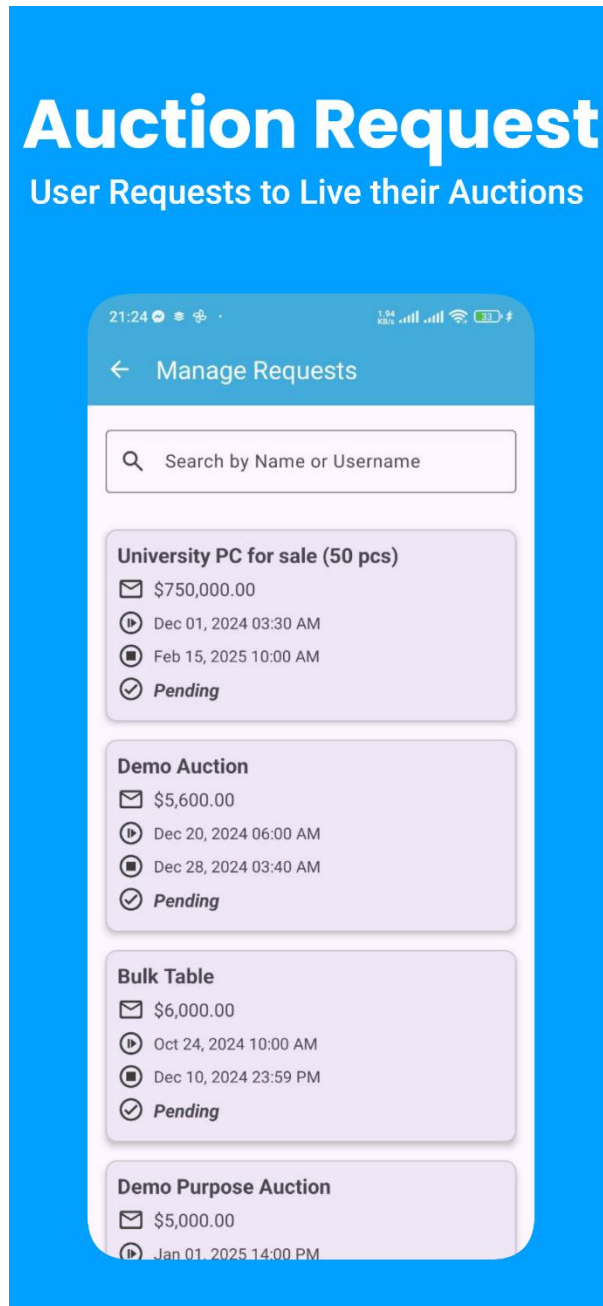


Figure 4.16: Manage Auction Request Activity

The Auction Request screen allows administrators to review and manage auction submissions. It displays details like item name, price, schedule, and status, ensuring easy navigation and decision-making for approving or rejecting requests.

4.3.5 Bids Info Activity

This is the Bids Info screen of the BidSell Admin Panel, enabling administrators to manage all user bids.

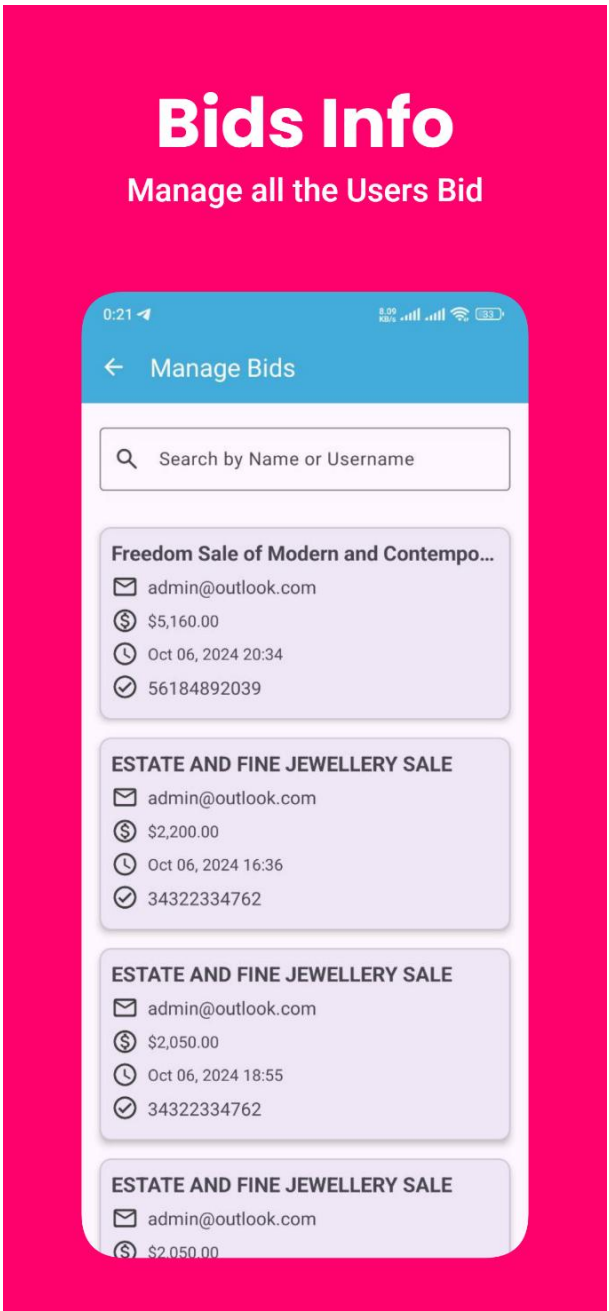


Figure 4.17: Bids Info Activity

The Bids Info screen provides tools for tracking and managing bids placed by users. It displays bid details, including the amount, user email, timestamp, and reference ID. The search bar allows quick navigation, while the organized layout ensures easy monitoring and efficient management of the bidding process.

4.3.6 Manage Transactions Activity

This is the Transactions Info screen of the BidSell Admin Panel, used to monitor and manage user transactions.

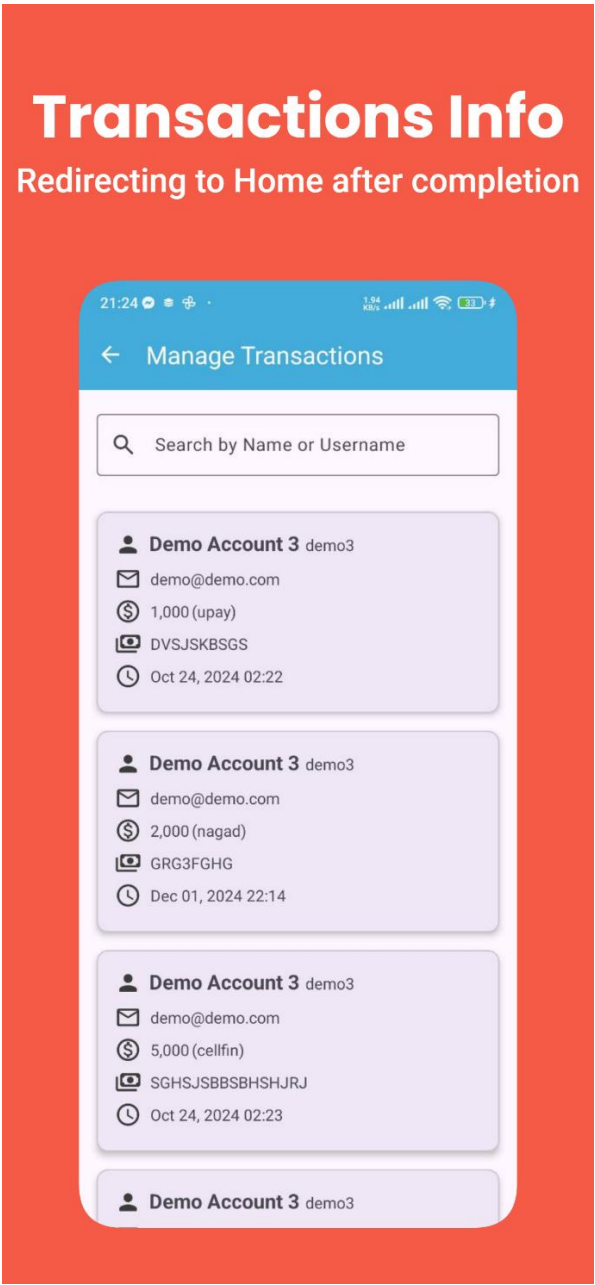


Figure 4.18: Manage Transactions Activity

The Transactions Info screen provides an overview of all user transactions. It displays details such as the user’s name, email, transaction amount, payment method, transaction ID, and timestamp. A search bar is included for quick navigation to specific transactions. This screen ensures efficient tracking and management of financial activities on the platform.

4.3.7 Auctions Info Activity

This is the Auctions Info screen of the BidSell Admin Panel, enabling administrators to manage all auctions.

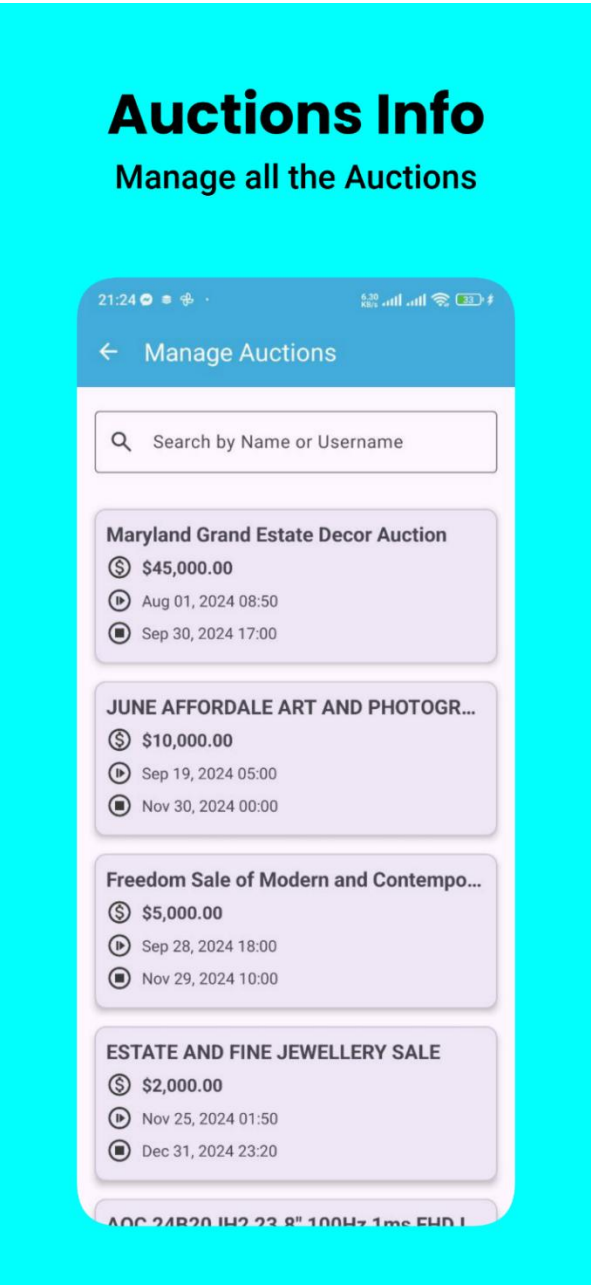


Figure 4.19: Auctions Info Activity

The Auctions Info screen provides administrators with tools to monitor and manage auctions. It displays key details such as auction title, starting price, start and end times. The search bar allows for quick access to specific auctions, while the organized layout ensures smooth navigation and efficient auction oversight.

4.3.8 Add Auction Activity

This is the Add Auction screen in the BidSell Admin Panel, allowing administrators to create new auctions.

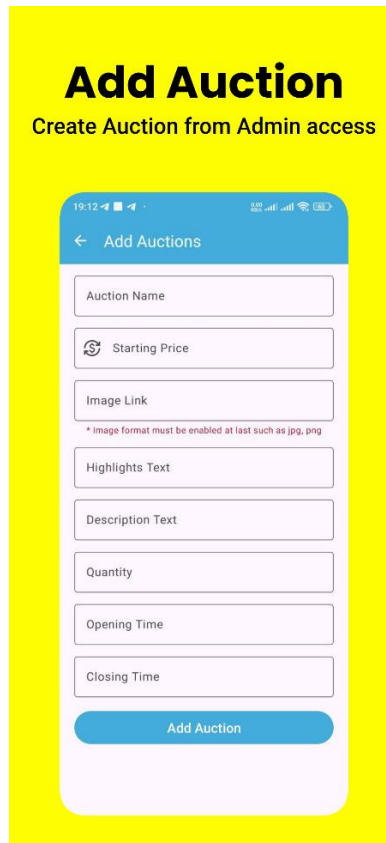


Figure 4.20: Add Auction Activity

The Add Auction screen enables administrators to create auctions directly. It includes input fields for auction details such as name, starting price, image link, highlights, description, quantity, opening time, and closing time. Once all fields are filled, the auction can be submitted using the Add Auction button. This feature ensures efficient and centralized auction management by administrators.

4.4 Conclusion

The implementation phase successfully brought the BidSell platform to life, integrating all planned features and modules. Key functionalities such as managing users, auctions, bids, transactions, and requests were developed with an intuitive and user-friendly interface. The admin dashboard provides a comprehensive overview of platform activities, ensuring effective management. The use of search filters, action buttons, and detailed views enhances usability, while the clean, organized design ensures seamless navigation and functionality. This phase establishes a robust and scalable foundation for the auction platform, ready to deliver a reliable and engaging user experience.

Chapter 5

Conclusion & Future work

5.1 Conclusion

The **BidSell** Project successfully achieved its objective of developing a secure, user-friendly, and dynamic online auction platform tailored to the specific needs of the Bangladeshi market. By employing the Agile methodology, the project embraced an iterative and flexible approach that ensured the platform's development remained aligned with user needs and evolving market demands. The platform addresses critical gaps in existing auction systems, such as the absence of user verification, real-time bidding, and transparency, by introducing innovative solutions like secure authentication, dynamic bidding, and intuitive user interfaces.

The system's architecture is designed around three distinct panels Buyer, Seller, and Admin that facilitate seamless user interactions and efficient management. Buyers can confidently browse and bid on valuable goods, while sellers can easily list and manage their auctions. The admin panel provides comprehensive tools for overseeing activities, ensuring compliance, and fostering trust among users. Key features, including category management, secure payment integration, enhance the platform's usability and functionality. The use of modern technologies, such as Firebase Authentication for secure login and Firebase Firestore for real-time database synchronization, ensures the platform's reliability, scalability, and security.

This project not only bridges the gap in user-friendly auction platforms in Bangladesh but also contributes significantly to the country's ongoing digital transformation. By creating a transparent, trustworthy, and accessible online marketplace, BidSell empowers individuals and businesses, driving economic growth and strengthening user confidence in digital transactions. The project has also provided invaluable learning opportunities for the development team, fostering expertise in software development, real-time systems, and e-commerce technologies.

Looking ahead, BidSell has identified several areas for enhancement, including the development of a responsive web application, integration of localized payment gateways, implementation of AI-driven fraud detection systems, and personalization of user experiences through machine learning algorithms. These future developments will further solidify BidSell's position as a transformative platform in the online auction industry. The successful completion of this project marks a significant milestone in addressing the needs of the digital commerce ecosystem in Bangladesh while showcasing the potential for innovation in this domain.

5.2 Limitation

While the BidSell platform offers a significant improvement over existing solutions, it has some limitations that need to be addressed in future iterations:

- **Platform Dependency:** The application is only available for Android, excluding iOS users and those who prefer desktop-based web applications.
- **Lack of Advanced Auction Features:** Features like automated bidding, predictive analytics, and detailed auction insights are yet to be implemented.
- **Limited Personalization:** The platform lacks advanced algorithms to offer personalized recommendations based on user behavior and preferences.
- **Language Support:** The app is currently only available in English, which may alienate users who prefer local languages like Bangla.
- **Fraud Prevention Mechanisms:** While the system includes basic user verification, it lacks AI-powered fraud detection to identify and mitigate potential risks effectively.
- **Notification System:** The notification system lacks specific alerts, such as reminders for upcoming auctions or immediate updates when a user's bid is outbid.

5.3 Future Work

As the platform evolves, several key areas of development have been identified to enhance its capabilities and reach:

1. Web Application Development:

- Expand **BidSell** into a responsive web-based platform to increase accessibility and cater to users who prefer desktop or browser-based interactions.

2. Enhanced Payment System:

- Introduce support for more diverse and localized payment gateways to streamline transactions and reduce processing times.
- Integrate automated invoicing and detailed transaction analytics for better financial tracking.

3. AI-Powered Fraud Detection:

- Develop AI-driven mechanisms to identify and mitigate fraudulent activities, ensuring a safe and trustworthy environment for users.

4. Advanced Auction Features:

- Implement tools like automated bidding systems, auction analytics, and real-time auction monitoring to improve functionality for both buyers and sellers.

5. Personalized User Experience:

- Add machine learning algorithms to offer personalized auction recommendations based on user behavior and preferences.
- Introduce multilingual support to make the platform more inclusive and accessible to a diverse user base.

6. Improved Control Systems:

- Strengthen admin tools for better monitoring and control, including advanced reporting and audit functionalities.
- Create a dashboard for sellers to track their auction performance and analytics in real-time.

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